

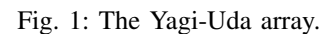
Abstract—In this report, a 6-element Yagi Uda antenna has been designed and analysed for an operating frequency of 433MHz and an input impedance of $50\ \Omega$ to send a signal approximately 150 meters to a receiving monopole. To optimize the parameters in the design of the Yagi-Uda, geometry, choice of material as well as other parameters were taken into consideration. The lengths and spacings of the directors and reflectors were optimized using simulations in COMSOL. The antenna has been characterized to have an impedance of $10.0 + 58.8j\Omega$ which is compensated with an L-network to $37.4 + 19j\Omega$. The simulations yielded a gain of 12.96 dB with an experimental signal strength of -67dBFS and 6.2dB.

Antennas are used to transmit and receive a form of electromagnetic radiation through the air with use of Electromagnetic waves. Electromagnetic waves can propagate information over long distances, hence their ubiquitous use for communication applications such as television, radio broadcasting, and satellite communications. Transmitter antennas transform bounded electrical signals into electromagnetic waves across the radio frequency band, which are then radiated into free space. Transmitter antennas can take a wide range of forms depending on specification, from simple monopole designs to more complex ones.

The report will begin with the theory section, which describes the basic principles behind the Yagi-Uda antenna and the theoretical background related to its performance.

¹Amplitude modulated

A Yagi-Uda can be simplified to three types of elements: the dipole, the reflector and the directors. The dipole, also referred to as the driving element, is the only powered component and can be seen in green in Fig. 1.



The phase interaction of each element with the EM field is dependent on the length of the element [5]. For a capacitor, the voltage lags the current, whereas for an inductor the voltage leads. The elements can be modelled as an LRC circuit, with their length representing the capacitive/inductive component value around the half-wavelength length, with no phase shift at that length. For the Yagi-Uda design [2], it is desired for the reflectors to be inductive, and directors to be capacitive such that the spacings needed for constructive and destructive

interference as seen in Fig. 2, can be minimized to be much shorter than a wavelength.

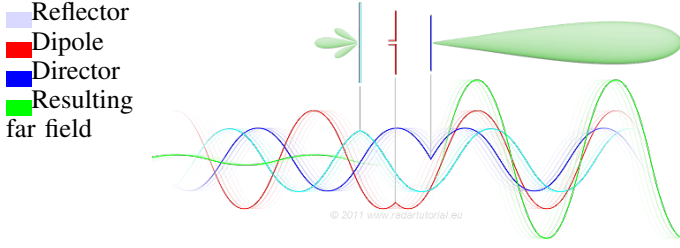


Fig. 2: Elements of the Yagi-Uda and the EM waves of the elements in their respective phases [6].

B. Directivity & Gain

The directivity of an antenna is the measurement of the concentration of a signal in a single direction, considering no losses relative to signal in all other directions. The directivity can be expressed as the ratio of transmitted power in the direction of propagation divided by the transmitted power by an isotropic antenna with a gain of 0dB, hence equal radiation in all directions. This results in equation 1 [7], where D_0 = directivity, U = radiation intensity along the considered direction, U_0 = the isotropic radiation intensity, and P_{rad} = the total radiated power.

$$D_0 = \frac{U}{U_0} = 4\pi \frac{U}{P_{rad}} \quad (1)$$

The gain G of an antenna describes the amount of power transmitted in the direction of propagation, taking into account the radiation efficiency e_r of the antenna. Consider equation 2[7].

$$G = e_r * D_0 \quad (2)$$

C. Analysis of two-element yagi-uda

The Yagi-Uda is a type of parasitic array², meaning the total field of the array is the vector addition of the fields radiated by all the individual elements [2]. Analysing the behaviour of a two-element Yagi-Uda, with one passive and one active element is necessary to be able to validate numerical approaches [3].

1) *Analytical solution:* The antenna can be considered as two infinitesimally thin dipoles, where the total field radiated is equal to the sum of the two elements, assuming no coupling between the elements [2]. In order to derive the electric and magnetic field components in the far field, the magnetic vector potential $\mathbf{A}$ can be found for the Hertzian dipole³.

$$A(\vec{r}, t) = \frac{\mu}{4\pi} l I_e(x', y', z') \frac{e^{-jkR}}{R} \hat{\mathbf{e}}_z. \quad (3)$$

²A type of endfire array, which contains a number of elements, only one of which is connected to a transmitter.

³Infinitesimally-thin straight wire of current

For an infinitesimally thin dipole, the current distribution is approximated in equation 4, which assumes that the antenna is center-fed and that the current vanishes at the end points.

$$I_1 = \begin{cases} I_0 \sin(k(\frac{l_1}{2} - z')) \hat{\mathbf{e}}_z & 0 \leq z' \leq \frac{l_1}{2} \\ I_0 \sin(k(\frac{l_1}{2} + z')) \hat{\mathbf{e}}_z & -\frac{l_1}{2} \leq z' \leq 0 \end{cases} \quad (4)$$

In the dipole, the current exists as a half-wave standing wave, where the time variation is irrelevant. On the other hand, the passive element located next to the dipole experiences an induced dipole due to the mutual coupling effect. To account for this, the current can be expressed to behave sinusoidally, with an additional term $\gamma e^{j\Psi}$. The term γ is implemented to account for the difference in magnitude, whereas $e^{j\Psi}$ is responsible for the phase difference [8]. This is affected implicitly by both the length and spacing between the two elements. Consider the current for the passive element in equation 5

$$I_2 = \begin{cases} I_0 \gamma e^{j\Psi} \sin(k(\frac{l_2}{2} - z')) \hat{\mathbf{e}}_z & 0 \leq z' \leq \frac{l_2}{2} \\ I_0 \gamma e^{j\Psi} \sin(k(\frac{l_2}{2} + z')) \hat{\mathbf{e}}_z & -\frac{l_2}{2} \leq z' \leq 0 \end{cases} \quad (5)$$

For an infinitesimal dipole of length dz' located on the z -axis, the electric field $\mathbf{E}$ and magnetic field $\mathbf{H}$ components in the far field are given in equation 6 and 7 [2]:

$$dE_\theta \approx j \frac{\eta k I_e(x', y', z') e^{-jkR}}{4\pi R} \sin(\theta) dz' \quad (6)$$

$$dH_\theta \approx \frac{dE_\theta}{\eta} \quad (7)$$

The fields are determined by integrating the current density I_1 and I_2 over the lengths of the dipoles, l_1 and l_2 , where, R is the distance to the observer from the dipole. To simplify the integral, two far-field approximations are made. First, the distance R located in the denominator is approximated to be the distance from observer to the centre of the dipole instead. This is known as the amplitude term [2]. Second, R is approximated to be $r - z' \cos(\theta)$ from the binomial expansion, where θ is the angle measured from the z -axis. This is known as the phase term. With the far-field approximations, the integral can be simplified even further by assuming equal lengths $l_1 = l_2 = l$ and evaluated as shown in seen in equations 8 and 9.

$$E_\theta \approx j \frac{\eta k e^{-jkR}}{4\pi r} \sin(\theta) \left[\int_{-\frac{l_1}{2}}^{\frac{l_1}{2}} I_1(x', y', z') e^{-jkz' \cos(\theta)} dz' + \int_{-\frac{l_2}{2}}^{\frac{l_2}{2}} I_2(x', y', z') e^{-jkz' \cos(\theta)} dz' \right] \quad (8)$$

$$E_\theta \approx j \frac{\eta k I_0 e^{-jkR}}{2\pi r} \left(\frac{\cos(\frac{kl}{2} \cos(\theta)) - \cos(\theta)}{\sin(\theta)} \right) (1 - \gamma e^{j\Psi}) \quad (9)$$

In equation 9, the term $(1 - \gamma e^{j\Psi})$ is considered the array factor. The array factor can be used to determine the total field of the array, which is equal to the field of a single element positioned at the origin multiplied by the array factor. Further analysis can be conducted to analytically determine time-average Poynting vector W_{av} [2], shown in equation 10.

$$\mathbf{W}_{av} = \frac{1}{2} \text{Re}[\mathbf{E} \times \mathbf{H}^*] = \hat{\mathbf{a}}_r \frac{1}{2\eta} |E_\theta|^2 \quad (10)$$

The Poynting vector is then used in equation 11 and 12 to determine the radiated power P_{rad} and the radiation intensity U .

$$P_{rad} = \oint \mathbf{W}_{av} \cdot d\mathbf{S} \quad (11)$$

$$U = r^2 W_{av} \quad (12)$$

Lastly, equation 11 and 12 are used to analytically determine the directivity D_0 as shown in equation 1 [2]. Performing the calculation becomes more difficult with more elements. Even though, the principle of superposition applies to variations in lengths and spacings alter the behaviour of mutual coupling effect needed for the induced current. As a result, the optimization of the Yagi-Uda will continue using numerical solutions in COMSOL, however the directivity for a two element Yagi-Uda will be compared analytically and numerically.

D. Signal Interface and Impedance Matching

1) *Impedance matching*: During signal transfer, the signal can suffer from adverse effects from incorrect characterisation of one mode. An effect most prevalent in an incorrectly characterised antenna is the reflection of the signal and the formation of standing waves back into the transmitter module.

To assess the magnitude of the reflected signal against the transmitted signal in a two-port network such as the antenna and transmitter, a scattering matrix (S-matrix) is analysed. The S-matrix is a way of interpreting the transmissions and reflections in a system as separate ratios between and within ports. The S-matrix for the two-port system shown in figure 3 can be seen in equation 13.



Fig. 3: Simplified schematic of a two port network

$$\begin{bmatrix} b_1 \\ b_2 \end{bmatrix} = \begin{bmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} \quad (13)$$

The port of interest is the connection between the transmitter module and the antenna, therefore, the S_{11} parameter is

considered, as it deals with reflection from the input to the antenna. This parameter is expressed in equation 14.

$$S_{11} = \frac{b_1}{a_1} = \frac{V_{reflected}}{V_{transmitted}} \quad (14)$$

Equation 14 in terms of voltage is also known as the reflection coefficient. This can also be represented in terms of the input impedance of the load (Z_L) and the output impedance of the transmitter module (Z_0) as seen in equation 15.

$$S_{11} = \frac{Z_L - Z_0}{Z_L + Z_0} \quad (15)$$

If the output impedance of the module and the input impedance of the antenna are correctly matched, the ratio of reflected transmitted power will be zero and no reflection will occur which reduces the losses of the antenna.

The reflection coefficient is also closely related to the standing wave ratio (SWR). When the signal is reflected, a standing wave is formed back into the transmitter, which interferes either constructively or destructively with the incoming wave. The ratio of the maximum versus the minimum of the amplitude due to the interference is then known as the SWR, with an ideal value of one.

2) *Use of a balun*: In a normal configuration, a dipole antenna connected to a coax cable would have one antenna arm attached to the inner and one attached to the outer conductor, wherein, it is grounded. For the inner conductor, the current does not have anywhere to flow apart from the antenna arm. In comparison, for the outer conductor, the current can either flow down the antenna arm or, in addition, may flow down the outside of the grounded outer conductor due to the skin effect in a phenomena known common mode currents. This results in unbalanced currents flowing down each antenna arm and, therefore, may lead to suboptimal antenna performance.

To correct the unbalanced behaviour, a balun may be used. Furthermore, due to the simplicity of construction, a folded balun will be explored. The construction of the folded balun consists of connecting the inner conductor of the coax cable to one antenna arm whilst connecting the same conductor at a half wavelength offset to the other antenna arm. Then by connecting the grounds of the coax cable together, the flows of current become oriented so as to cancel out these common mode currents.

This is an important addition to the antenna for the reduction of outside interference.

III. DESIGN

Before a simulation of a Yagi can be developed, an initial design alongside some reasonable range of values need to be specified to condense the numerical computation. The simulation limitations will yield additional design constraints.

A. Constraints

1) *Material Choices*: For an arbitrary Yagi-Uda array, elements with a larger conductivity can yield a larger gain. However, conductivities higher than $10^7 [\text{S/m}]$ lead to a negligible

increase in gain [9]. The materials available for construction with conductivity higher than the aforementioned threshold are copper and aluminium [10]. Aluminium was chosen due to the lower cost and larger range of radii available.

2) *Number of Directors:* To analyse the impact of the number of directors on the gain, the gain has been computed for a range of directors using the Antenna toolbox in MATLAB [11]. As shown in Fig. 4, the gain increases with the number of directors before starting to slowly fall off. The plot also shows the total length of material needed for the directors. It can be concluded that 4 directors will approximately yield a gain of 9.4dB, which should be sufficient to transmit a signal across a distance of 150 meters, assuming the antenna has high efficiency.

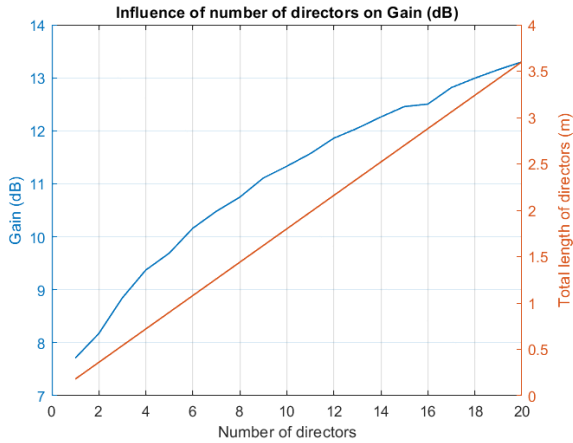


Fig. 4: Effect of Number of Directors on the Gain (dB) produced by a Yagi-Uda antenna for 433MHz. Produced using Antennas Toolbox [11]

To ensure a sufficient signal at the intended distance of 150m away from the antenna, a value of at least 8 dBi was chosen as the aim for antenna directivity, alongside a maximum length of material based on budget. From this, the number of directors was chosen to be four.

3) *Radius of elements:* High-frequency signals create a surface current density as most current flows along the surface (skin) of the conductor. The minimum radius required is defined as skin depth, being equal to $\sqrt{\frac{\rho}{\pi f \mu}}$, where, ρ is the resistivity of the material, μ the permeability and f the frequency of the current. This results in a skin depth of $3.94 \times 10^{-6} [m]$ [7]

Taking into account the availability and reduced cost per unit length, a cylindrical rod of 8mm is to be used for the build, with an increased radius decreasing the power losses due to heat of the driver.

4) *The Boom:* For strength and simplicity in adjustment, a design making use of a conductive boom was utilized. The directors are located a distance of 4cm above and parallel to the plane of the boom. The parallel arrangement results in a

negligible impact on peak forwards directivity ($< 0.1dB$), and a slight impact on the overall shape of the radiation pattern in all other directions according to situations, as can be seen in Appendix A Fig. 13.

The effects of a conductive boom such as the one in this design can be compensated with slightly modified director lengths [4]. For cylindrical or square booms, the increase in element length is linearly proportional to $\frac{d_{cylinder}}{\lambda}$. The proportionality translates to a difference in director length of $0.002\lambda \approx 1.3mm$ from expectation to compensate for the effects of the boom. Due to the very small impact from both simulation and literature, this effect will be neglected in the design.

B. Element Lengths

1) *Driving element and Reflector:* A half-wave dipole or a half-wave folded dipole antenna is commonly used as the driving element of a Yagi-Uda [12]. The length of the driving element ranges from 0.45λ to 0.49λ instead of $\lambda/2$ to achieve resonance, with the aim to reduce the imaginary part of the input impedance to zero, so that the reactance will vanish once the antenna is impedance matched [2]. Due to this fact, the length of the driving element has been chosen to be 0.47λ .

Alongside the driving element, the length of the reflector must also be defined. Based on the experimental findings of Peter P. Vezbickie[4], the optimal reflector length is 0.482λ . It results, in the desired interaction between the EM waves returning towards the dipole interact and the newly outgoing waves.

2) *Directors of uniform length:* Traditionally, the lengths of the director elements are taken to be equal and generally smaller than the length of the driving element. Alternatively, an optimization algorithm can be deployed to set directors to varying lengths.

It has been experimentally shown [4] that the optimal uniform director lengths for a maximum directivity depend on the total number of directors and the diameter of the elements. The elements of smaller and larger diameters will have the same maximum gain, where the optimal length of the prior would be larger than those of the former. Given that the element diameter is 8mm and there is a total of 4 directors, the initial uniform length is 0.425λ [4].

3) *Directors of varying lengths:* Various numeric methods [5], [13], [14] have been developed to assign a unique length to each director by optimizing for directivity, front-to-back ratio and reduction of side lobes.

Jones *et al.* made use of a genetic algorithm (GA)[14] that works on the premise of evolution and propagates the generations of those designs that are optimal for the defined parameters. Alternatively, gradient descend methods[5], [13] have been successfully used by Cheng *et al.* and Adekola *et al.*

The gradient descent method searches the local maxima in a smaller subset defined by the initial estimation and the defined range, while GA performs an effective global search. The performance of a design is evaluated by an approximation

of the current induced into the passive elements via a method of moments (MoM)[13], [14] or an approximation with a three-term expansion with complex coefficients[5].

A literature overview of improvements reachable through these techniques is summarized in table I.

TABLE I: Optimization[5], [13] of directors' length while keeping a uniform spacing.

	Front-to-back ratio (dB)	Side lobes (dB)	Directivity* (dB)
Cheng <i>et al.</i> Initial estimation	-	-	8.77
Cheng <i>et al.</i> optimized	-	-	10.0 (32%)
Adekola <i>et al.</i> uniform lengths	5.2331	-5.222	-
Adekola <i>et al.</i> optimized lengths	7.1152 (26%)	-6.998 (25%)	-

C. Element Spacing

1) *Uniform/non-uniform directors spacing*: Unique spacing between each director can likewise be found through GA[13], [14] or gradient descent[5] algorithms and yield significantly better results than arrays with uniform spacing. Results demonstrating the possible improvements through these methods are summaries in table II.

TABLE II: Optimisation[5], [13] of spacing between directors' while keeping lengths uniform.

	Front-to-back ratio (dB)	Side lobes (dB)	Directivity ⁴ (dB)
Cheng <i>et al.</i> Initial	-	-	8.77
Cheng <i>et al.</i> optimized	-	-	10.68 (55%)
Adekola <i>et al.</i> uniform spacing	5.2331	-5.222	-
Adekola <i>et al.</i> optimized	8.45005 (38%)	-8.447 (38.2%)	-

2) *Reflector and driving element spacing*: The impact of the spacings between the dipole and reflector as well as the first director can be considered for a 2-element Yagi-Uda set up, described in the theory section. By examining the plots of the radiation patterns, the differences between the two types of passive elements can be visualized for a variety of distances to the driving element. To model the passive elements, the director is set to a length 10% shorter, while the reflector is 10% longer than the driving element, as shown in the theory section [12]. Consider Fig 5, for the length of a director.

As indicated in Fig. 1, the reflector is located on the left of the driving element, whereas the director is placed on the right. Taking a closer look at the directivity plot, for instance, 0.7λ , it can be seen that the radiated power is directed to the right. This supports the argument, that the distance between the directors and the driving element should be less than

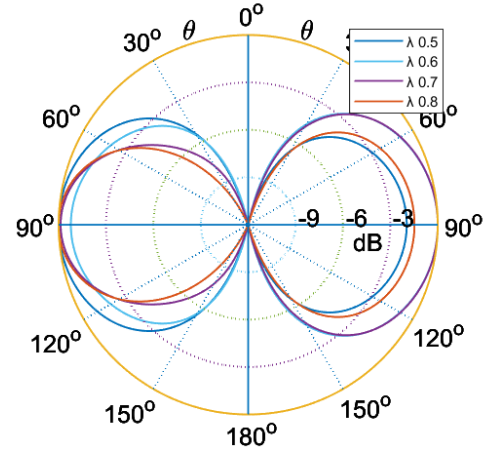


Fig. 5: Radiation plot of 10 % shorter passive element for a range of spacings and dipole length of 0.47λ [11]

the distance between the reflector and the driving element. Consider, the radiation plot for the longer passive element in Fig. 6.

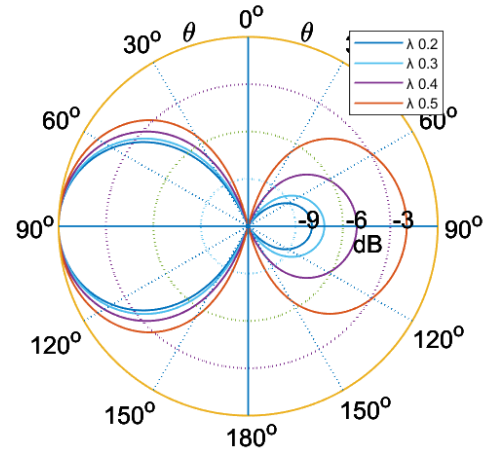


Fig. 6: Radiation plot of 10 % longer passive element for a range of spacings and dipole length of 0.47λ [11]

It can be concluded that the reflector behaves in opposition to the director. Shorter spacings of $0.3-0.4\lambda$ are preferred to maximize the directivity leftwards. Using these values, the optimization in COMSOL will be limited ranges from $0.2-0.3\lambda$ for the reflector spacing and $0.5-0.7\lambda$ for the director.

D. Initial design

When a simultaneous optimization with all element lengths and spacings being independent is performed, the maximum

attainable gain is increased significantly as found from the literature, review *et al.*[5] and Jones *et al.*[14]. Keeping in mind the literature results, the final set of parameters for the initial design can be condensed into table as seen in Appendix A, Table V.

The directors are taken to be of a uniform length of 0.425λ . This corresponds to the experimentally found[4] optimal length for an array of 6 elements, four being directors, of $8mm$ in diameter. The length of the reflector and the driving element, as well as the distance between them, are chosen to be 0.482λ , 0.47λ and 0.2λ respectively. These values are based on literature values from Peter P. Viezbicke[4], as well as constraints and evaluations made in the design section. The range of values for which the simulation will optimize for is described in the element spacing, and a range of $0.2-0.5\lambda$.

E. Design of the Matching Circuit

The impedance of an antenna was simulated, yet due to the target optimized variable being directivity, this value was offset from the desired $50\Omega + 0j$. To account or the difference, a matching network made up of an LC circuit was designed to match the output impedance of the transmitter. A central concept in developing a matching circuit is a smith chart, as can be seen in figure 7.

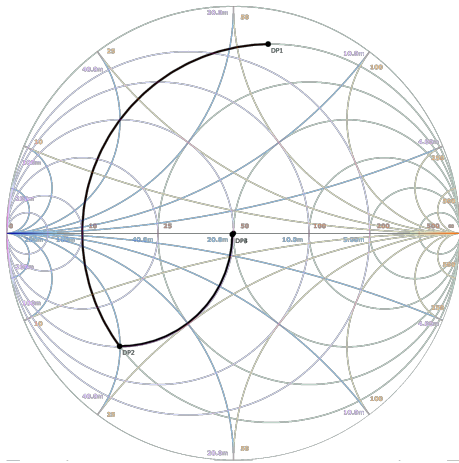


Fig. 7: Smith chart for impedance matching, with use case as applied to final design

The chart is used to visualize the effect of inductors and capacitors on the input impedance of a simple LC circuit. The movement along the admittance (blue) or impedance (green) represents a component placed in parallel or series respectively, with the component type and its value determining the distance and the direction of movement along the lines.

IV. OPTIMISATION OF ELEMENT SPACINGS AND ELEMENT LENGTHS

The initial design was been optimized with regard to literature and basic simulations in the previous section of the report. In the optimization section, the initial design will be built upon

in terms of spacings and lengths of directors, alongside the validation of the model.

A. COMSOL Simulation Environment Design and Validation

A well-constructed and validated simulation environment is essential to worthwhile numerical analysis. Symmetry is often essential in reducing the complexity of a system, yet in this case this is not possible due to far-field behaviour being of interest and the antenna producing a distinct radiation pattern in all three dimensions. The far-field requirement also yields the definition of the outer boundary layer being a far-field domain, which by means of radiation impedance matching allows for a much less memory-intensive representation of infinite space.

All materials were set to the in-built versions of those used in the real construction, including aluminium elements and a steel boom for later tests. The boom was implemented in later tests to verify the presumed negligible effect from literature [4].

1) *Comparison to Analytic Solutions:* As a form of validation, a 2-element Yagi-Uda was constructed in the COMSOL simulation environment to allow for a comparison with the analytical solution described in the theory. The model consists of a half-wave dipole with a lumped port at its center-point, and a finite length rod located a distance of 0.2λ away. To make for more data points in the comparison, the simulation and analytical analysis has been conducted for a range of three lengths, where the length of the passive element varied by 10 % while keeping the dipole constant.

A visual comparison can be seen in Fig. 14, in Appendix A, where the radiation plot of the **E** and **H** are shown side by side. The analytical approach was visualized using the the antenna toolbox in MATLAB [11], which uses equation 9 to determine the directivity. On the other side, the COMSOL solution is shown, which has been computed for a fine mesh size and a significantly larger far-field. Although, some variation between the two methods can be seen, both planes show similarities in general shape and, more importantly, the forward and backward lobes.

For a more in-depth comparison, consider Table. III, showing the numerical differences.

Passive element length	gain, analytical (dB)	gain, numerical (dB)	Difference
0.5λ	6.50	6.4030	0.097
0.6λ	4.67	4.8205	0.150
0.4λ	4.45	4.8106	0.360

TABLE III: Maximum gain of 2-element Yagi-Uda, solved numerically and analytically.

A difference in values is present, it can be said that the approximations made during in the analytical approach as well as the numerical, still have an impact on the results. These results are sufficiently accurate, and allow use to continue.

2) *Mesh scaling*: With the analytical solution validating the model, the converging behaviour of mesh size is also tested. A course mesh allows for fewer degrees of freedom to be simulated, hence significantly decreasing computational load. The most relevant parameters for antenna performance were evaluated for convergence for mesh scaling factor sweep from 0.5 to 3.0 seen in table IV. The data reveals a limited impact on the value of directivity with minimal convergence, hence justifying the use of a 0.5 mesh scaling factor for bulk simulations, considering directivity is the parameter for which the design will be optimized. There is greater deviation for the impedance value, therefore a finer mesh simulation will be used to determine the final simulated impedance value for a certain design.

Mesh Scale	Directivity (dB)	Impedance (Ω)
0.5	9.522	46.84
1.5	9.475	45.10
3.0	9.433	44.6

TABLE IV: Antenna performance parameters as a function of mesh size showing convergence

B. Far-Field Matching Layer Size

By definition, the impedance matched boundary layer must encompass the entire geometry of the tested body, which results in a very large volume within. Minimizing the enclosed volume could greatly reduce the number of degrees of freedom, hence greatly decreasing simulation time. The effect of decreasing the surrounding volume was analysed using an additional parameter sweep for a constant antenna geometry, which yielded insignificant impact on the final data, but a very rapidly increasing simulation time even for slightly larger radii. A plot of the found effect can be seen in Appendix A, Fig 15.

C. COMSOL Optimization Approach

As was determined previously, spacing and length values obtained from literature were often fairly unpredictable and without a simplified trend or mathematical expression describing the optimal arrangement [2], [14]. This finding meant that due to the lack of a pruning algorithm in COMSOL, an iterative search of optimal array designs within a smaller range of length and spacing variation was performed. This approach can only be assumed appropriate if a sufficiently good design is used as an initial estimation, which will act as a centre point for further iteration within the range.

In order to keep the simulation feasible and the computation time down to a manageable level, a step size of 0.1λ was used for both spacings and lengths. Fig.16 in Appendix A, shows a radiation plot for all the combinations of lengths and spacings considered in the optimization. The optimal value is the right most red outline, of which the parameters are shown in table VI, in Appendix A.

The optimized antenna can be seen in Fig 8 showing the radiation behaviour in 3D, with a maximum gain of 10.2 dB. The multiple side lobes present indicate a uniform coverage of destructive interference sideways, with each lobe having low

gain compared to the forwards lobe. The final simulation also yielded a lumped port impedance of $27.676 + 23.912j$.

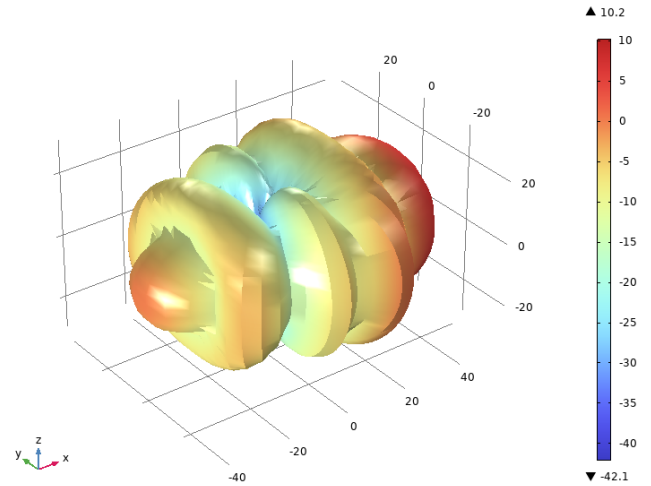


Fig. 8: 3D directivity plot of the simulation of the final design

V. CHARACTERISATION OF A YAGI-UDA ANTENNA

The built antenna can be seen in Fig. 9. Once built, the impedance of the antenna was characterized and matched, which allowed for further testing of the design.



Fig. 9: Image of the experimental setup

A. Impedance and Matching

By mounting the antenna on the stand such that interference with the ground would be minimal, the *miniVNA Tiny* with VNA software would be used to measure the impedance of the antenna without a matching circuit, yielding an impedance of $10.0 + 58.8j\Omega$. In line with the Smith chart of Fig. 7 the antenna was matched to output the impedance of the transmitter, yielding the circuit seen in Fig. 10 with corresponding component values.

Repeating the characterization but for the matched circuit in line with the antenna yielded the following VNA software plot in Fig. 11 showing an impedance with a real component

of $R = 37.4\Omega$ and a complex component of $X = 19j\Omega$. Additionally, the plot also yields the SWR value of 1.69 showing that a large portion of the signal propagates effectively through the boundary between the transmitter module and the antenna.

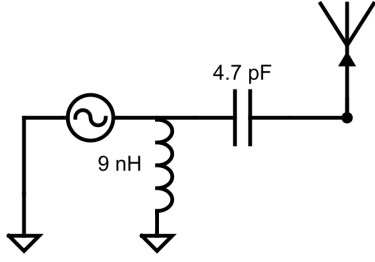


Fig. 10: Impedance matching circuit

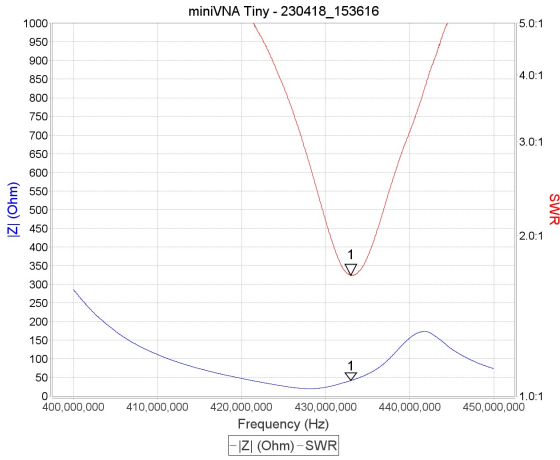


Fig. 11: Impedance and standing wave ratio measured from matched antenna

B. Characterization of gain

To characterize the gain of the antenna, a measurement was conducted using the Airspy receiver software and a receiving antenna. The antenna was oriented such that the polarization of the transmitter and receiver 20.0m away were in line, and was then rotated to yield the directivity plot in Fig. 12

The signal strength was measured at multiple positions around the antenna, with extra measurements at the main lobe. The Airspy program measures in dBFS, which is a measure of the ratio of the maximum power of the signal versus noise. The measurements were performed 20 meters from the antenna such that the values correspond well within the far field boundary of $\frac{2d^2}{\lambda}$, with d being the largest dimension of the antenna. The final measurement was then performed in the intended range of 150 meters which yielded a dBFS of -67 .

VI. CONCLUSION

The goal of this project was to design and make an antenna that can send a signal to reach a dipole at a distance of 150m

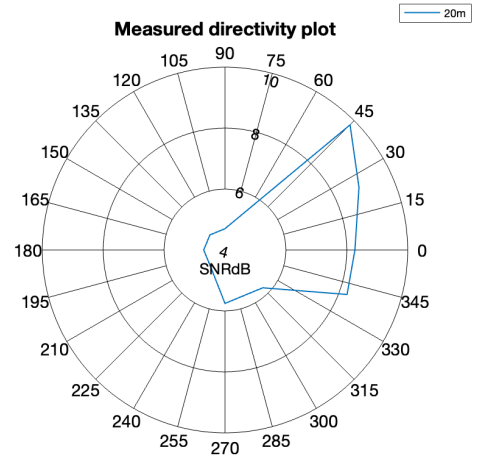


Fig. 12: 2D plot of signal strength to noise ratio (SNRdB) measured experimentally, higher values correspond to a stronger signal

at a frequency of 433MHz. Using the design and optimization process shown in this report, the 6-element Yagi-Uda with the parameters shown in table VI was able to successfully meet the goal with a gain -67 dBFS.

An indicative final measurement was the directivity plot showing a highly lopsided result with a directivity of only 6.2dB towards 0 deg, yet 9.2dB when pointed towards 45 deg. From the only asymmetrical element of the antenna being the boom, it is likely that the simulations for this misrepresented the reflective behaviour which the boom had on the directionality of the signal, likely as a result of a simplified boom geometry. Ground effects could additionally contribute to this effect, yet were not taken into account during the measurements or simulations. This effect of the boom could be easily negated by using a non-conductive material instead. Regardless of the boom, the front to back ratio of the antenna was excellent, which proves the potential for this antenna to perform well if the boom was replaced.

In terms of characterization, the antenna was found to have an impedance of $37.4 + 19j\Omega$ instead of the required $50.0 + 0j\Omega$ from the transmitter. This loss of signal strength significantly impacted the performance of the antenna, as reflected by the 1.69 SWR. Beyond a more closely tuned matching circuit, and environmental conditions during characterization, the impedance could have been optimized for as opposed to the directionality based approach of the report.

The final antenna was sufficient to reach the desired specifications, yet due to an insufficiently detailed modelling of the boom in the design, this effect ended up dominating the final lack in performance of the antenna. all other simulation steps were verified and yielded expected results both for approximate directivity, impedance and gain.

VII. ACKNOWLEDGMENT

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VIII. APPENDIX

A. Additional Figures and tables

Lengths						
	l1	l2	l2	l4	l5	l6
Distance (λ)	0.482	0.47	0.425	0.425	0.425	0.425
Optimization range (λ)	N/A	N/A	0.2-0.5	0.2-0.5	0.2-0.5	0.2-0.5
Spacings						
	d12	d23	d34	d45	d56	
Distance (λ)	0.25	0.6	0.3	0.3	0.3	
Optimization range (λ)	0.2-0.3	0.5-0.7	0.2-0.5	0.2-0.5	0.2-0.5	

TABLE V: Initial design parameters

Lengths						
	l1	l2	l2	l4	l5	l6
Distance (λ)	0.482	0.47	0.37	0.36	0.37	0.35
Spacings						
	d12	d23	d34	d45	d56	
Distance (λ)	0.25	0.6	0.32	0.45	0.35	

TABLE VI: Optimized design parameters

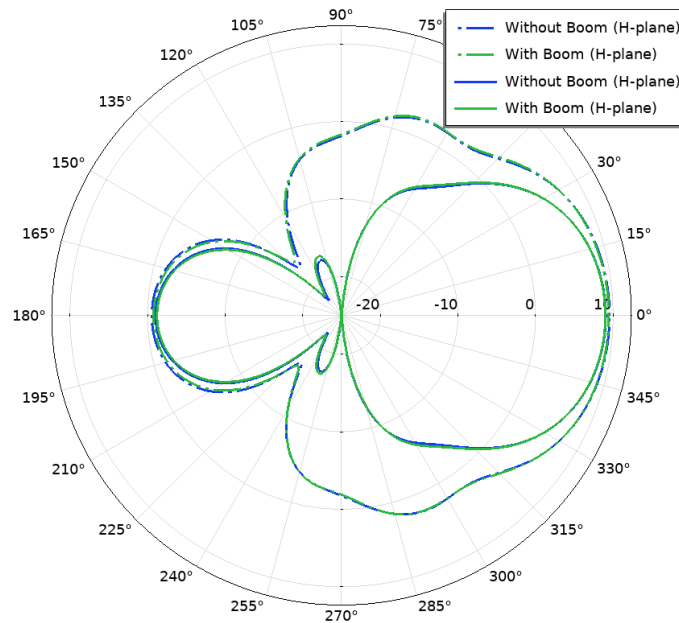


Fig. 13: Radiation pattern plot showing the effect of the boom, for a non ideal arrangement of spacings and lengths

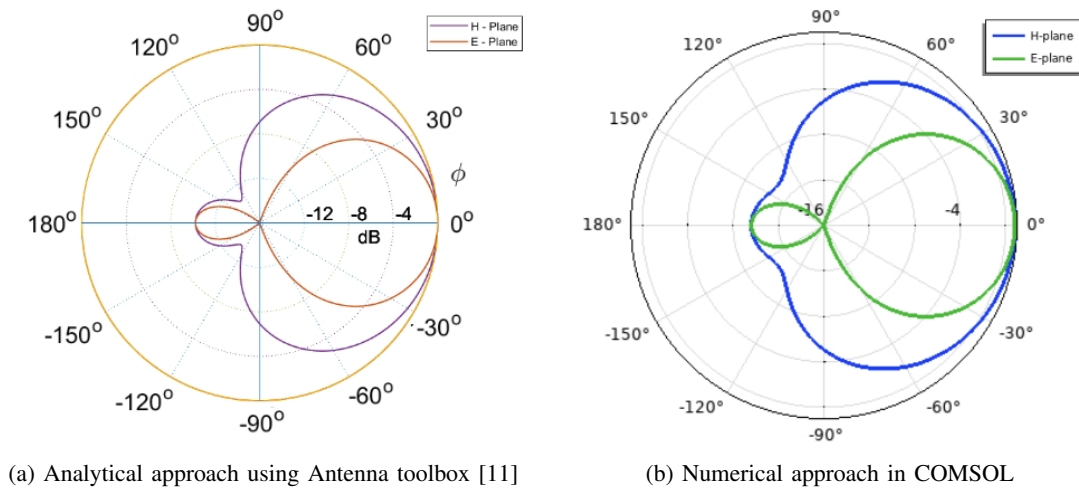


Fig. 14: Radiation plot for 2-element Yagi-Uda

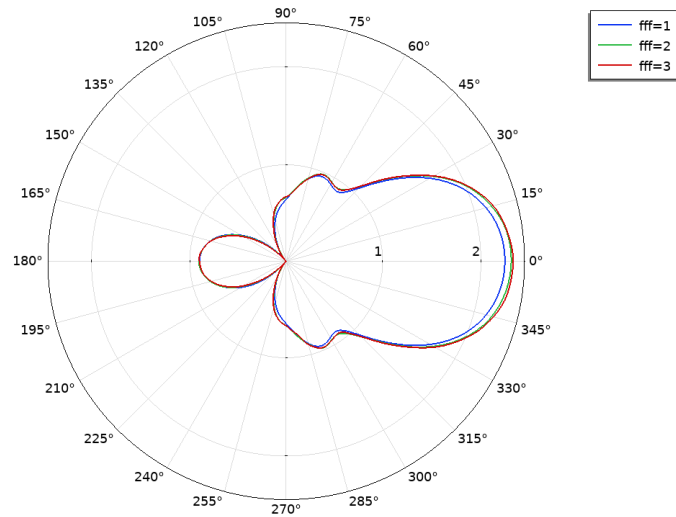


Fig. 15: Sweep of the radius of the impedance matched boundary far-field region showing no impact on results as a function of radius

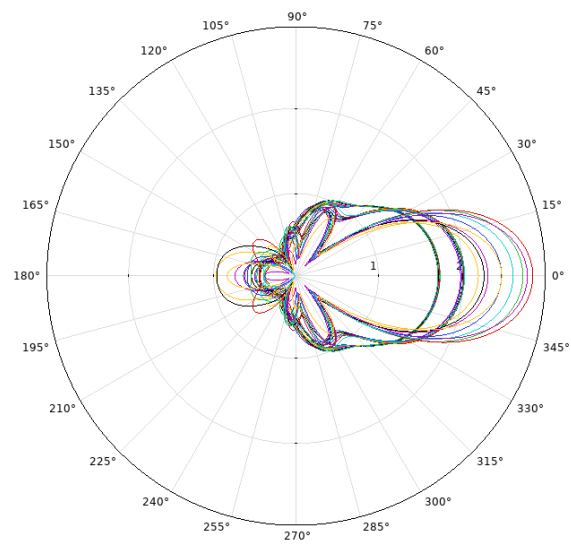


Fig. 16: Parameter-Sweep showing all the lengths and spacing combinations ran in COMSOL, in a directivity plot.